

CIS 6270

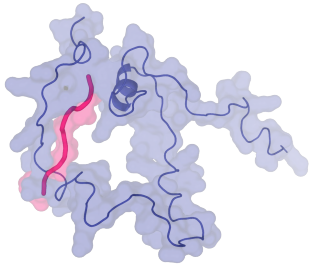
Module 4: Discrete Diffusion

Lecture 4.3: Reverse Rates, Guidance, and the Rate-Matching Loss

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Pranam Chatterjee, PhD
Assistant Professor of Bioengineering
Assistant Professor of Computer and Information Science
University of Pennsylvania

Penn Engineering
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At stationarity, a process in detailed balance looks the same run forward or backward in time, and we call it reversible.

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Probability circulates around the cycle. Run backward in time, it would circulate the other way.

Let’s see both pictures side by side

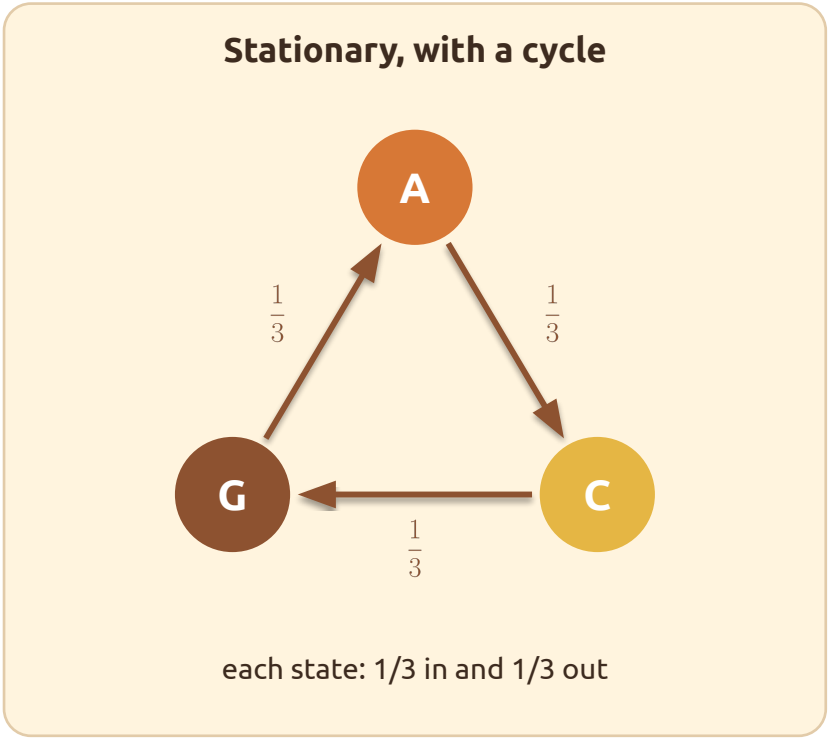
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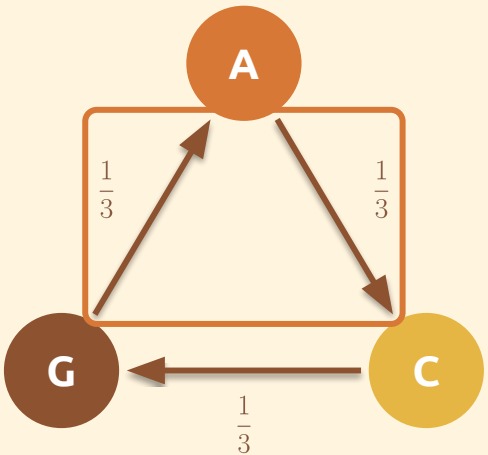
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On the left, probability moves around a cycle of three bases



At each state, 1/3 arrives and 1/3 leaves, so the uniform distribution is stationary

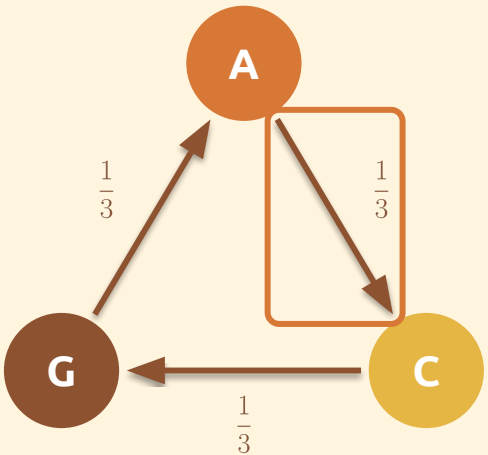
Stationary, with a cycle



each state: 1/3 in and 1/3 out

The flux from A to C is 1/3, and nothing flows back

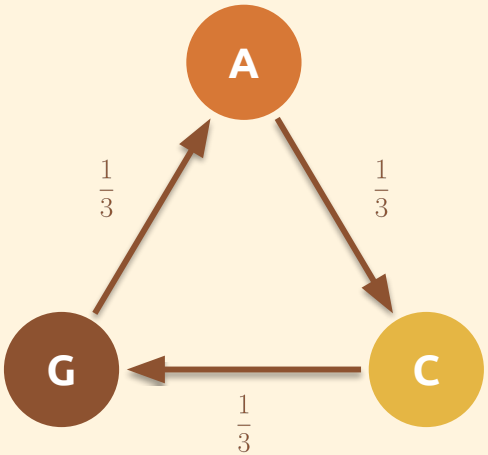
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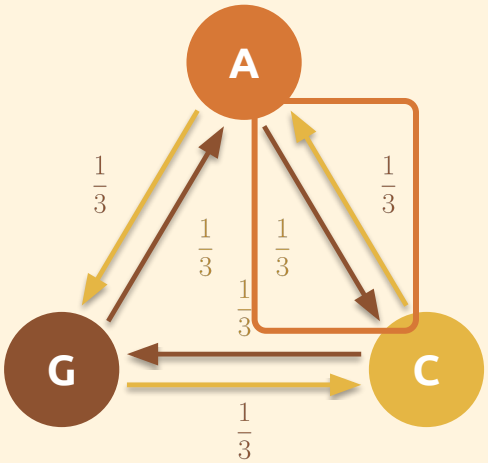
In detailed balance, the flux is equal in both directions along every edge

Stationary, with a cycle



each state: 1/3 in and 1/3 out

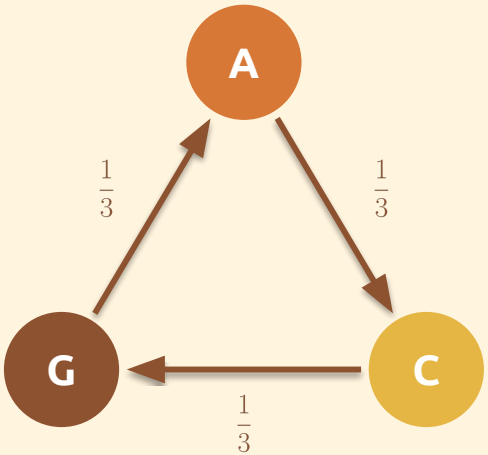
Detailed balance



each edge: equal flux both ways

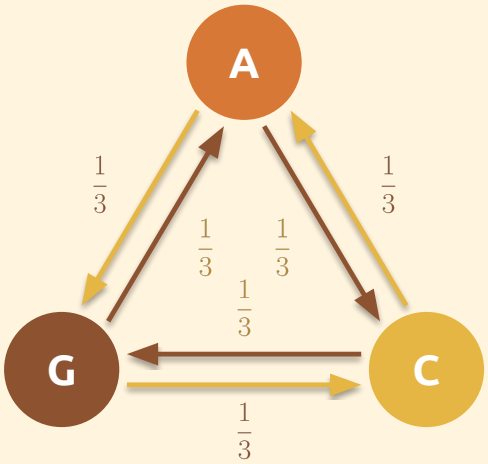
At stationarity the flux balances at each state, and in detailed balance it balances on each edge

Stationary, with a cycle



each state: 1/3 in and 1/3 out

Detailed balance



each edge: equal flux both ways

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For uniform π, every rate between two distinct tokens equals 1/K, so

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For uniform π over K tokens, the flux $\pi(x)R_\pi(x, y)$ from x to y equals 1/K times 1/K, which is also the flux $\pi(y)R_\pi(y, x)$ from y back to x.

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Masking reaches its stationary distribution through one-way jumps, so its reverse process can only reveal tokens.

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In Lecture 3, to reverse an SDE we needed the score $\nabla \log p_t$. Let's find what we need for reversing jumps.

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This is detailed balance between the forward process and its reversal.

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From Bayes' rule, we get the same answer. For $x \neq y$,

$$\Pr\left(X_t = x \mid X_{t+h} = y\right) = \frac{h\,R_t(x,y)\,p_t(x)}{p_t(y)} + o(h)$$

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The probability that the process was at x at time t, given that it is at y at time t + h, equals h times $R_t(x, y)$ times $p_t(x)$ divided by $p_t(y)$, up to o(h).

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The coefficient of h is the reverse rate, and the probability ratio enters through Bayes' rule.

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If x is twice as probable as y, the reverse jump from y to x must run twice as fast as the forward jump.

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Substitute the reverse rates into the master equation:

$$\sum_{x \neq y} \left[p_t(x) \hat{R}_t(x, y) - p_t(y) \hat{R}_t(y, x) \right] = -\frac{d}{dt} p_t(y)$$

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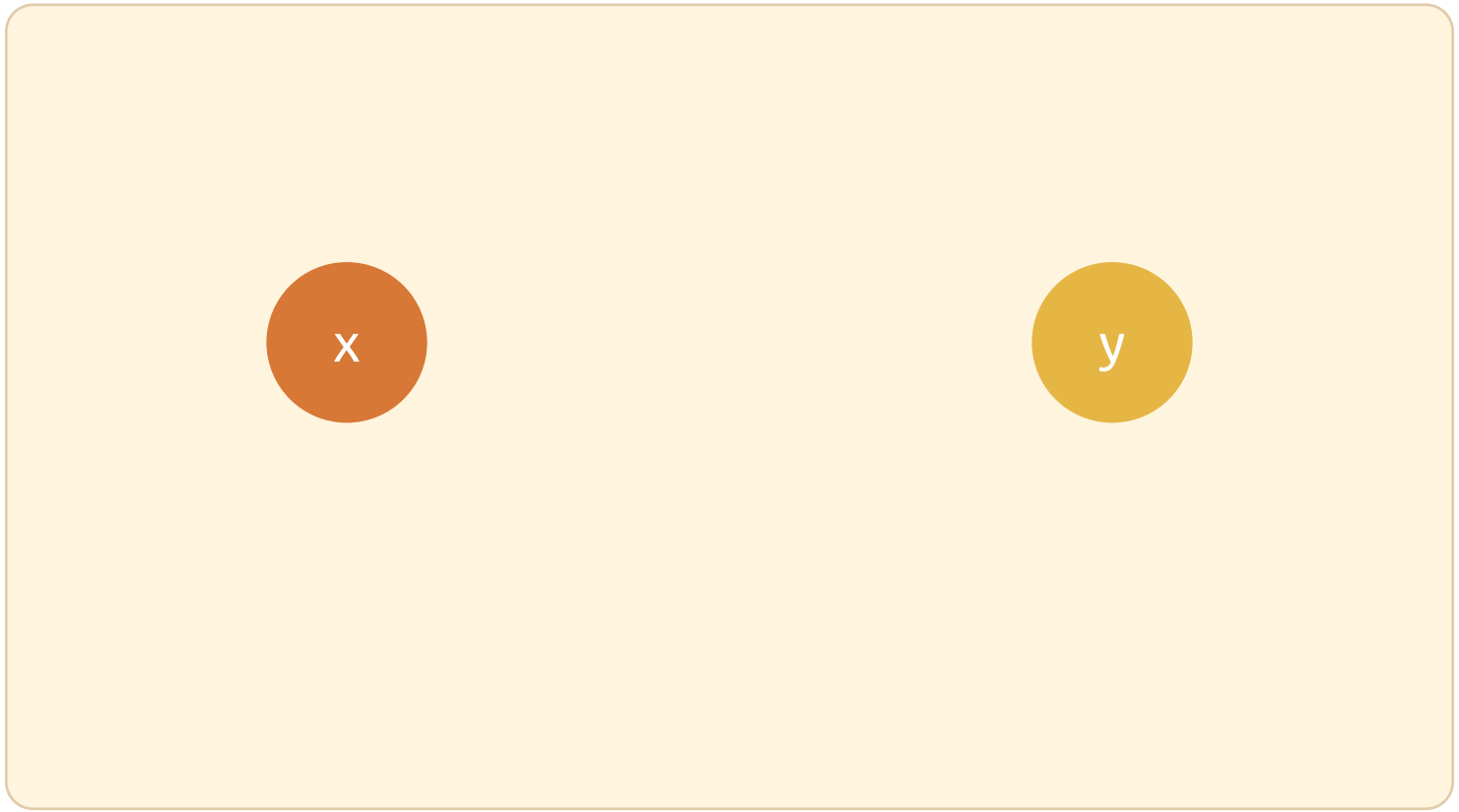
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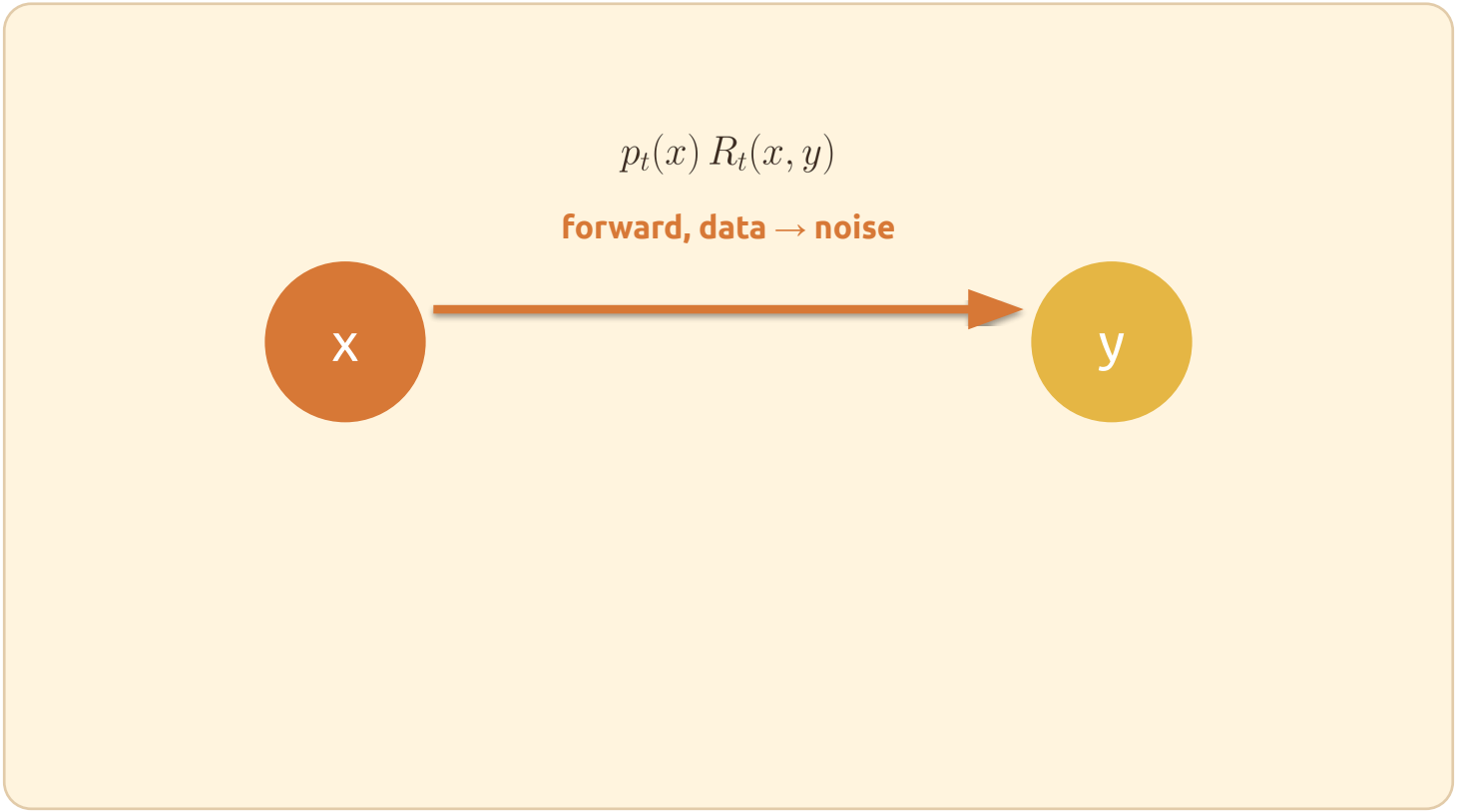
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If we run the reverse rates from p_1 back to time zero, we pass through every marginal p_t and end at the data distribution.

Let’s look at one edge of the state space



In the forward process, flux $p_t(x)R_t(x, y)$ moves from x to y



In the reverse process, the same flux must go from y to x

$$p_t(x)\, R_t(x, y)$$

forward, data → noise



reverse, noise → data

$$p_t(y)\, \hat{R}_t(y, x)$$

Match them edge by edge to get the reverse rate

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So the reverse process retraces p_t from noise back to data

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The forward process changes one position at a time, so we only need ratios between sequences that differ at one position:

$$\boxed{s_t(z)_{i,a} = \frac{p_t(z^{i \leftarrow a})}{p_t(z)}}$$

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Each sequence has L(K – 1) such ratios, so a DNA sequence of length 10 has 30.

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Each ratio measures how a one-letter change affects the sequence's probability.

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The probability $q_t(z \mid x_0)$ of the noisy sequence z given the clean sequence x_0 equals the product over positions i from 1 to L of $q_t(z^i \mid x_0^i)$, the probability of the noisy token z^i given the clean token x_0^i.

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We corrupt every position independently, so $$q_t(z \mid x_0) = \prod_{i=1}^{L} q_t(z^i \mid x_0^i)$$

The conditioned process has the same rates, so its reversal follows the same rule:

$$\hat{R}_t(y, x \mid x_0) = R_t(x, y) \frac{q_t(x \mid x_0)}{q_t(y \mid x_0)}$$

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The conditional reverse rate $\hat{R}_t(y, x \mid x_0)$ from y to x, given the clean sequence x_0, equals the forward rate $R_t(x, y)$ from x to y times $q_t(x \mid x_0)$, the probability of x given x_0, divided by $q_t(y \mid x_0)$, the probability of y given x_0.

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During generation we no longer have x_0, so we need to connect these conditional rates to the marginal ones.

Write the marginal as a sum over clean sequences

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The marginal averages the conditional marginals over the data:

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The marginal probability $p_t(x)$ of the noisy state x equals the sum over clean sequences x_0 of the data probability $p_{data}(x_0)$ times the conditional probability $q_t(x \mid x_0)$.

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Divide by $p_t(y)$, then multiply and divide each term by $q_t(y \mid x_0)$:

$$\frac{p_t(x)}{p_t(y)} = \sum_{x_0} \underbrace{\frac{p_{\text{data}}(x_0)\, q_t(y \mid x_0)}{p_t(y)}}_{p(x_0 \mid X_t = y)} \frac{q_t(x \mid x_0)}{q_t(y \mid x_0)}$$

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The ratio $p_t(x)/p_t(y)$ equals the sum over clean sequences x_0 of the posterior $p(x_0 \mid X_t = y)$, which is $p_{data}(x_0)$ times $q_t(y \mid x_0)$ divided by $p_t(y)$, times the conditional ratio $q_t(x \mid x_0)/q_t(y \mid x_0)$.

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Multiply both sides by $R_t(x, y)$:

$$\boxed{\hat{R}_t(y, x) = \sum_{x_0} p(x_0 \mid X_t = y)\; \hat{R}_t(y, x \mid x_0)}$$

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The marginal reverse rate $\hat{R}_t(y, x)$ is the sum over clean sequences x_0 of the posterior $p(x_0 \mid X_t = y)$ of x_0 given y, times the conditional reverse rate $\hat{R}_t(y, x \mid x_0)$.

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Several clean sequences can explain the same noisy sequence, so each contributes to the reverse rate according to its plausibility.

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From the conditional construction, we get one target for each clean sequence, and with the posterior we combine those targets into one reverse rate at every state and time.

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From the conditional construction, we get one target for each clean sequence, and with the posterior we combine those targets into one reverse rate at every state and time.

This is the discrete counterpart of the marginal velocity from Lecture 2, which averaged conditional velocities over the same kind of posterior.

Let’s check that average on one base

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Let the clean base be A with probability 0.75 and C with probability 0.25, and suppose we observe C.

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Let the clean base be A with probability 0.75 and C with probability 0.25, and suppose we observe C.

For the two clean bases, we get two very different targets:

$$\hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{A}) = 1 \cdot \frac{0.625}{0.125} = 5, \qquad \hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{C}) = 1 \cdot \frac{0.125}{0.625} = 0.2$$

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If the clean base is A, the conditional reverse rate $\hat{R}_t(C, A \mid A)$ equals the forward rate 1 times 0.625 divided by 0.125, which is 5, and if it is C, $\hat{R}_t(C, A \mid C)$ equals 1 times 0.125 divided by 0.625, which is 0.2.

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Now weigh the two with Bayes’ rule:

$$p(\mathrm{A} \mid \mathrm{C}) = \frac{0.75 \times 0.125}{0.75 \times 0.125 + 0.25 \times 0.625} = 0.375$$

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The posterior probability p(A | C) that the clean base is A, given that we observe C, equals 0.75 times 0.125 divided by the sum of 0.75 times 0.125 and 0.25 times 0.625, which is 0.375.

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The marginal reverse rate is their average: $$\hat{R}_t(\mathrm{C}, \mathrm{A}) = 0.375 \times 5 + 0.625 \times 0.2 = 2$$

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For the two clean bases, we get two very different targets:

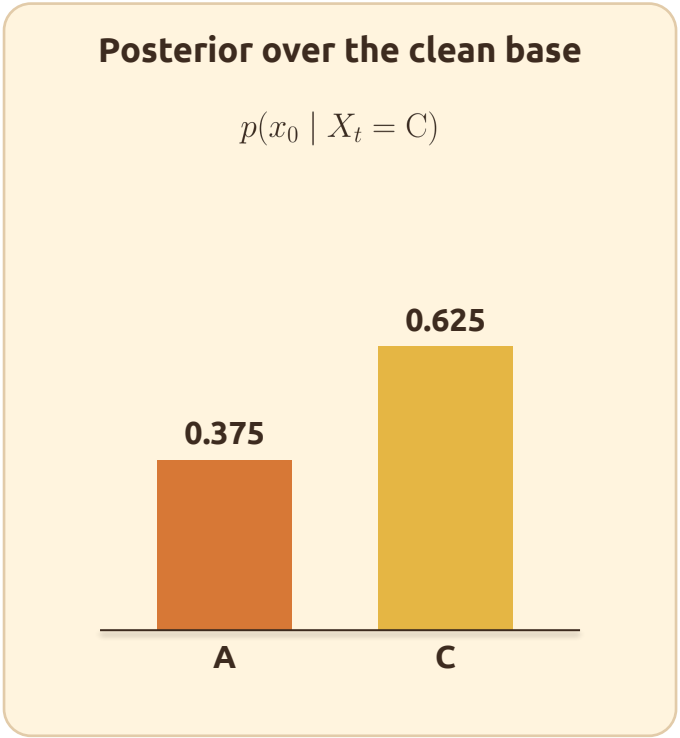
$$\hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{A}) = 1 \cdot \frac{0.625}{0.125} = 5, \qquad \hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{C}) = 1 \cdot \frac{0.125}{0.625} = 0.2$$

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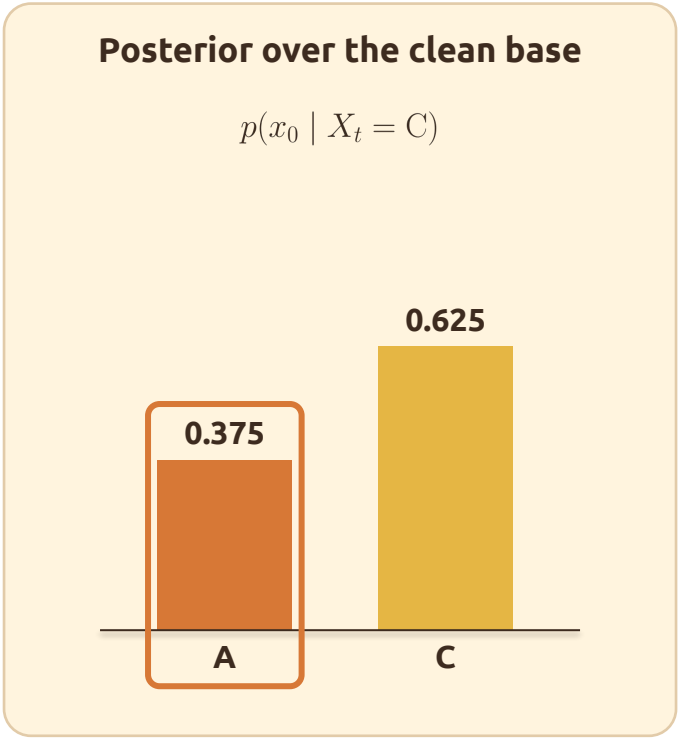
The marginal reverse rate is their average: $$\hat{R}_t(\mathrm{C}, \mathrm{A}) = 0.375 \times 5 + 0.625 \times 0.2 = 2$$

The marginal reverse rate $\hat{R}_t$(C, A) equals 0.375 times 5 plus 0.625 times 0.2, which is 2.

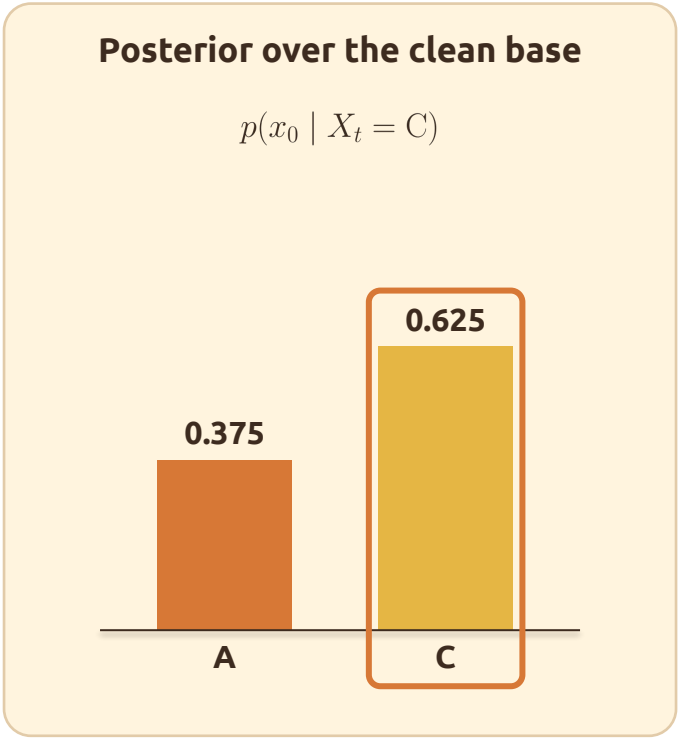
Two clean bases could explain the C we observed



By Bayes' rule, A has weight 0.375



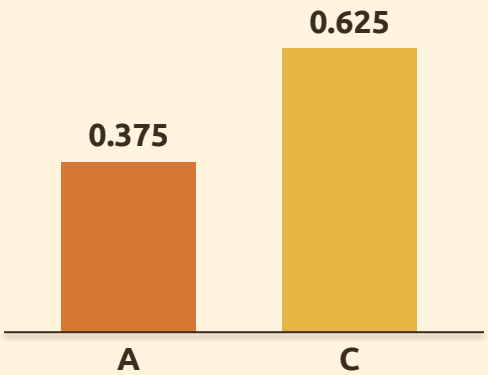
A clean C explains the observation better, so it gets weight 0.625



If the clean base was A, the reverse jump toward A runs at rate 5

Posterior over the clean base

$$p(x_0 \mid X_t = \mathrm{C})$$



Reverse rates from C toward A

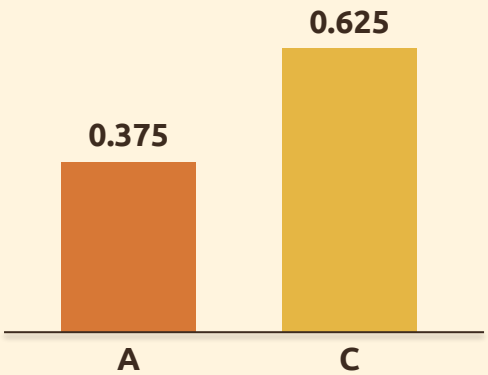
$$\hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{A}) = 5$$



If it was C, that same jump runs at rate 0.2

Posterior over the clean base

$$p(x_0 \mid X_t = \mathrm{C})$$



Reverse rates from C toward A

$$\hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{A}) = 5$$



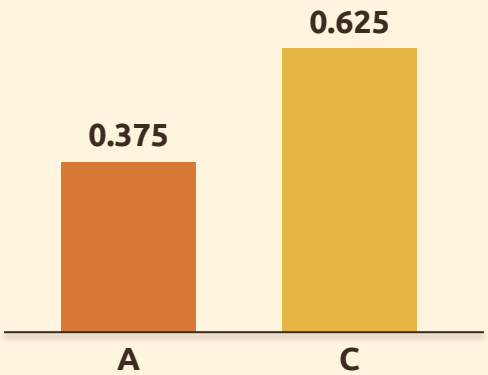
$$\hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{C}) = 0.2$$



The marginal reverse rate is the posterior average of the two, which is 2

Posterior over the clean base

$$p(x_0 \mid X_t = \mathrm{C})$$



Reverse rates from C toward A

$$\hat{R}_t(\mathrm{C}, \mathrm{A} \mid \mathrm{A}) = 5$$



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$$\hat{R}_t(\mathrm{C}, \mathrm{A}) = 2$$

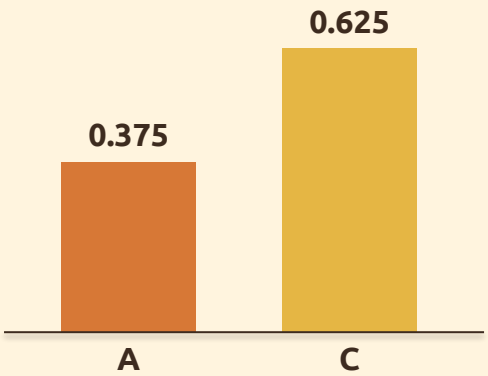


$$0.375 \times 5 + 0.625 \times 0.2 = 2$$

Checking directly, $p_t(A)/p_t(C) = 0.5/0.25 = 2$

Posterior over the clean base

$$p(x_0 \mid X_t = \mathrm{C})$$



Reverse rates from C toward A

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Now reduce the posterior to one position at a time

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The posterior over whole sequences is far too large to store, and the forward process only changes one position at a time.

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Change position i of z to token a. Every other factor cancels:

$$\frac{q_t(z^{i \leftarrow a} \mid x_0)}{q_t(z \mid x_0)} = \frac{q_t(a \mid x_0^i)}{q_t(z^i \mid x_0^i)}$$

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The ratio of $q_t(z^{i \leftarrow a} \mid x_0)$ to $q_t(z \mid x_0)$ equals $q_t(a \mid x_0^i)$ divided by $q_t(z^i \mid x_0^i)$, because the factor of every other position is the same in both.

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So for the marginal ratio, we need only the posterior of the clean token at position i:

$$\boxed{\frac{p_t(z^{i\leftarrow a})}{p_t(z)} = \sum_b p(x_0^i = b \mid z)\, \frac{q_t(a \mid b)}{q_t(z^i \mid b)}}$$

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The ratio of $p_t(z^{i\leftarrow a})$ to $p_t(z)$, for changing position i of z to token a, equals the sum over clean tokens b of the posterior probability $p(x_0^i = b \mid z)$ that position i was b, times $q_t(a \mid b)$ divided by $q_t(z^i \mid b)$.

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The denoiser $d_\theta^i(b \mid z, t)$ predicts this posterior from the whole noisy sequence.

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The network reads the whole noisy sequence, even though each rate changes only one position.

We also want to steer generation toward a property

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By Bayes' rule, we reweight each jump by how much more likely E becomes after it:

$$\Pr\left(X_{t+h} = y \mid X_t = x,\, E\right) = \Pr\left(X_{t+h} = y \mid X_t = x\right) \frac{h_{t+h}(y)}{h_t(x)}$$

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The probability of moving from x to y during a short step h, given the event E, equals the probability of that move without E times $h_{t+h}(y)$ divided by $h_t(x)$.

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Read off the coefficient of h:

$$\boxed{R_t^h(x, y) = R_t(x, y)\, \frac{h_t(y)}{h_t(x)}}$$

This is Doob's h-transform

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The function h itself satisfies an equation that we solve backward from the final time.

Compare h at two times, one short step apart

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This is the Kolmogorov backward equation, and with it the conditioned transition probabilities sum to one.

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The chance of reaching the goal now must match the average chance after one jump.

Now return to the reverse rates

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Detailed balance with π means that the forward rate $R_t(x, y)$ equals the rate $R_t(y, x)$ in the other direction times π(y) divided by π(x).

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Substituting it into the reverse rate, we get $\hat{R}_t(y, x)$ as $R_t(y, x)$ times $p_t(x)/\pi(x)$ divided by $p_t(y)/\pi(y)$.

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Define $h_t = p_t/\pi$:
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The reverse rate $\hat{R}_t(y, x)$ equals the forward rate $R_t(y, x)$ from y to x times $h_t(x)$ divided by $h_t(y)$, where the value h_t is the marginal p_t divided by the stationary distribution π.

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So generation is an h-transform of the noise process, reweighted toward states that are more likely under the data than under pure noise.

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The conditional reverse rate $\hat{R}_t(y, x \mid x_0)$ equals the forward rate $R_t(y, x)$ from y to x times $q_t(x \mid x_0)$ divided by $q_t(y \mid x_0)$, the probabilities of x and y given the clean token x_0.

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In our example, the jump from C to A is 5 times faster than noise when the clean base is A, and 5 times slower when it is C.

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The marginal reverse process averages these h-transforms with the posterior weights, so generation is a posterior mixture of h-transforms.

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The guided reverse rate $\hat{R}^c_t(y, x)$ from y to x equals the unguided reverse rate $\hat{R}_t(y, x)$ times $p(c \mid X_t = x)$, the probability of the property c from x, divided by $p(c \mid X_t = y)$, the probability of c from y.

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In Discrete Guidance (ICLR 2025), we estimate p(c | x) with a classifier on noisy sequences, and raise the ratio to a power γ to strengthen the guidance.

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The diffusion model stays fixed, and with guidance we only reweight its reverse rates.

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Suppose the classifier probabilities are $p(c \mid \cdot) = 0.2$ at C, 0.1 at A, 0.6 at G, and 0.2 at T.

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Multiply each rate by the ratio of classifier probabilities:

$$\hat{R}^{c}(\mathrm{C}, \cdot) = \left(2 \cdot \frac{0.1}{0.2},\ \ 0.5 \cdot \frac{0.6}{0.2},\ \ 0.5 \cdot \frac{0.2}{0.2}\right) = (1,\ 1.5,\ 0.5)$$

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For the guided rates out of C, we multiply the unguided rates 2, 0.5, and 0.5 toward A, G, and T by the classifier ratios 0.1/0.2, 0.6/0.2, and 0.2/0.2, and get 1, 1.5, and 0.5.

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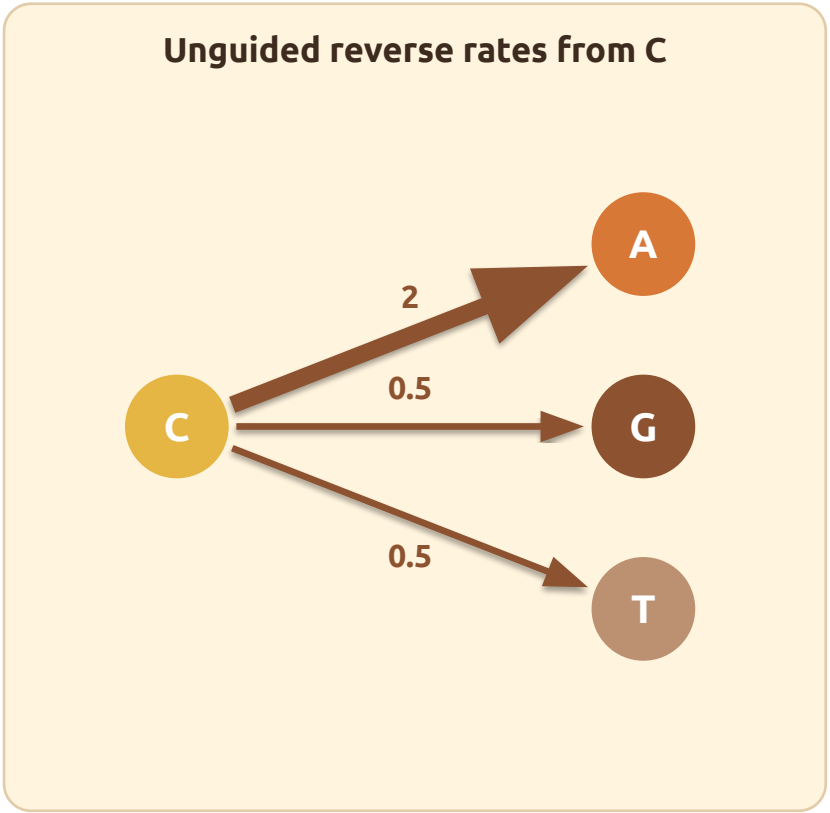
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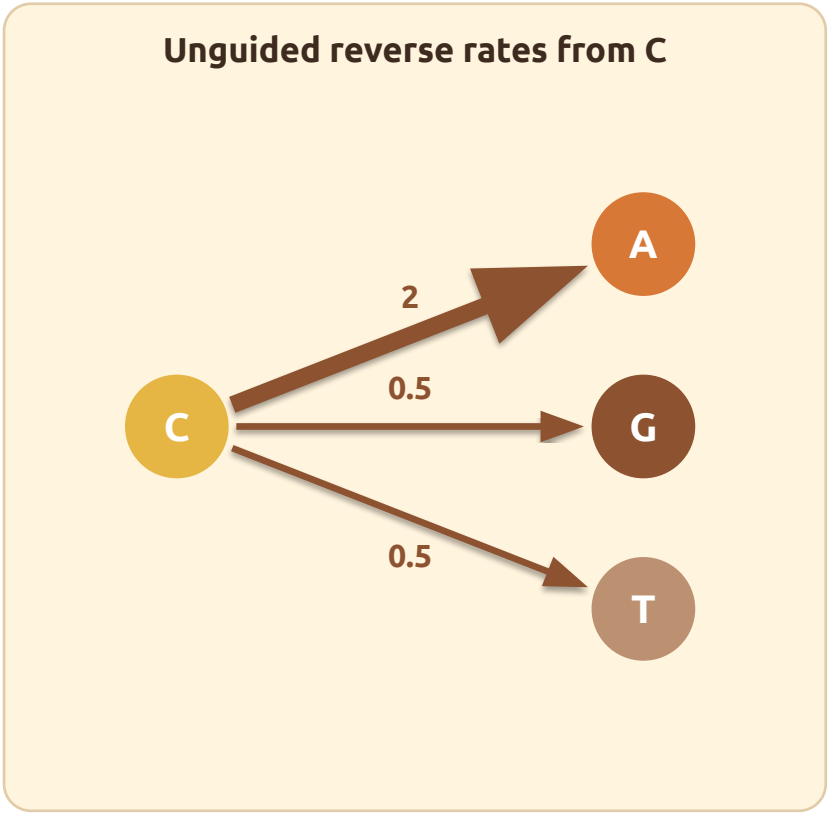
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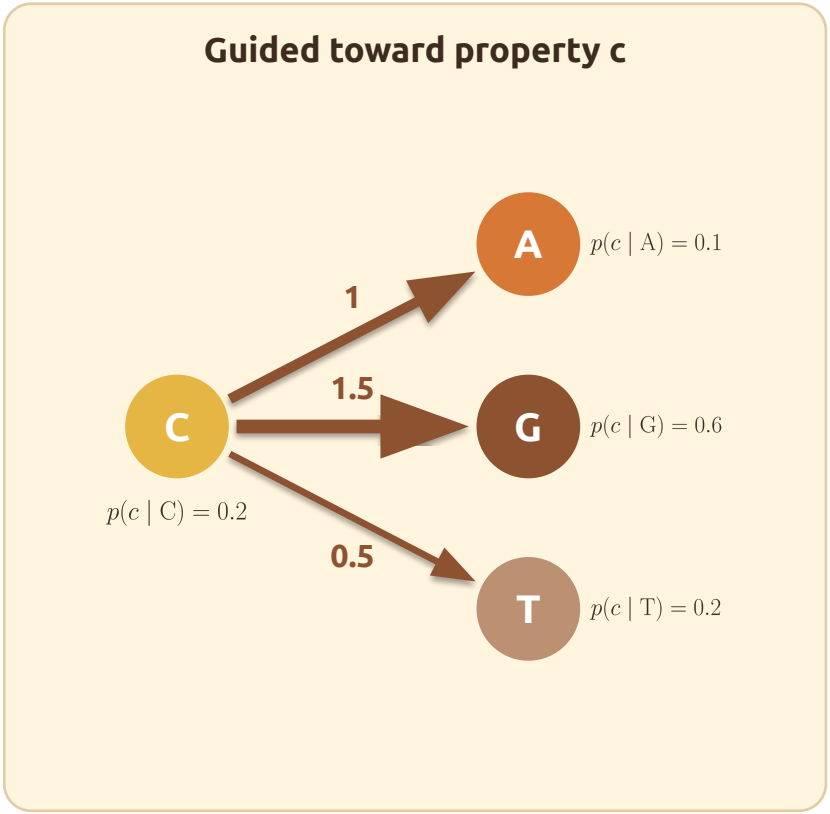
G becomes the most likely next base, and the jump toward A slows down, because A makes the property less likely.

Our three unguided rates out of C

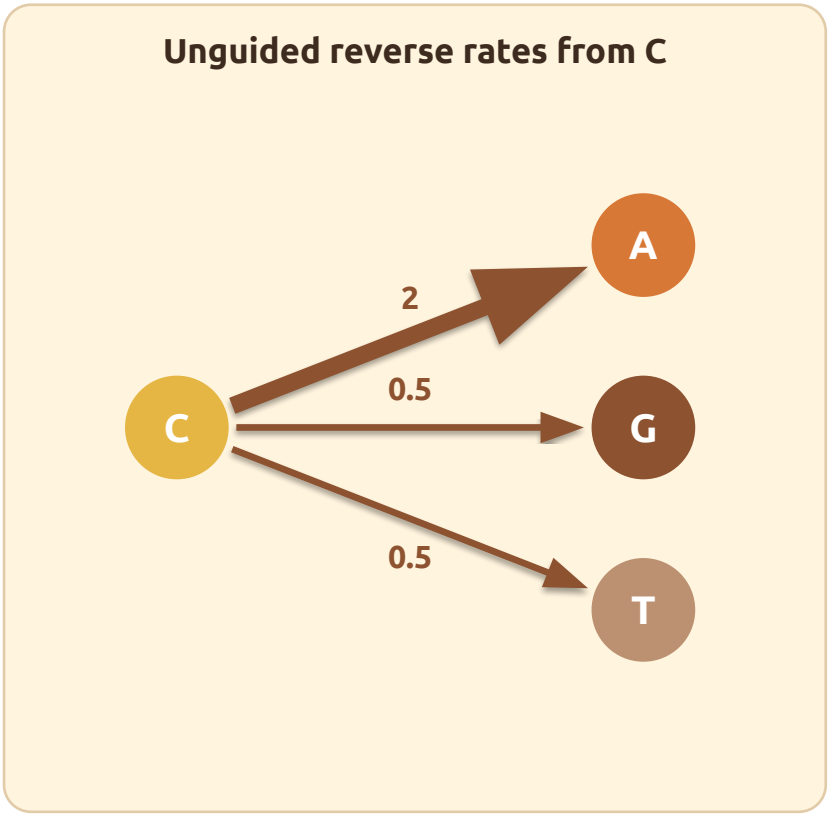


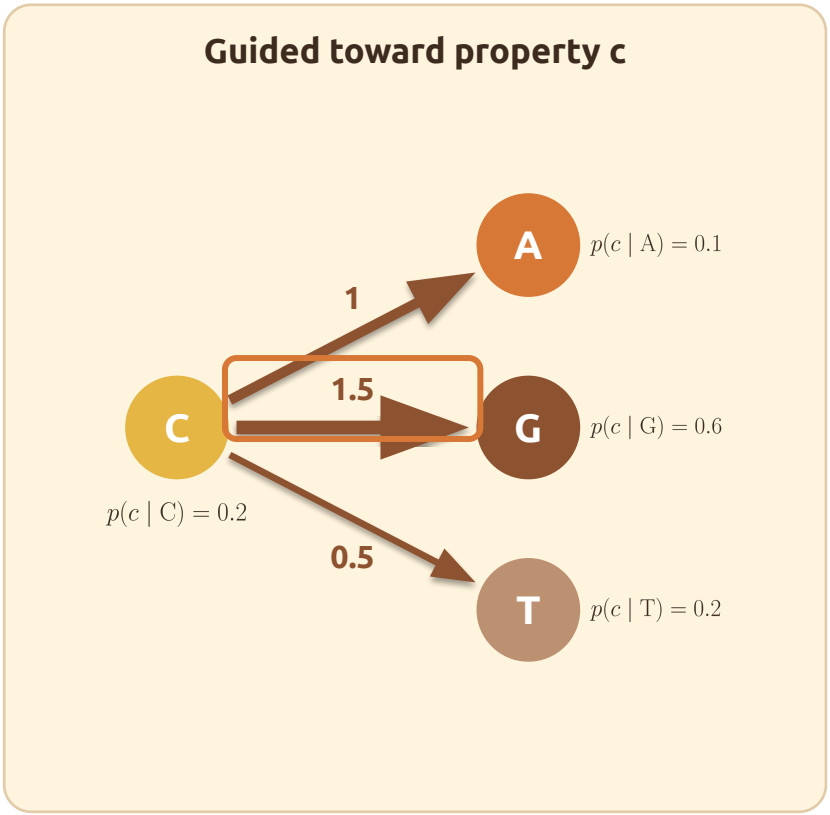
With the classifier, we score each base for the property c



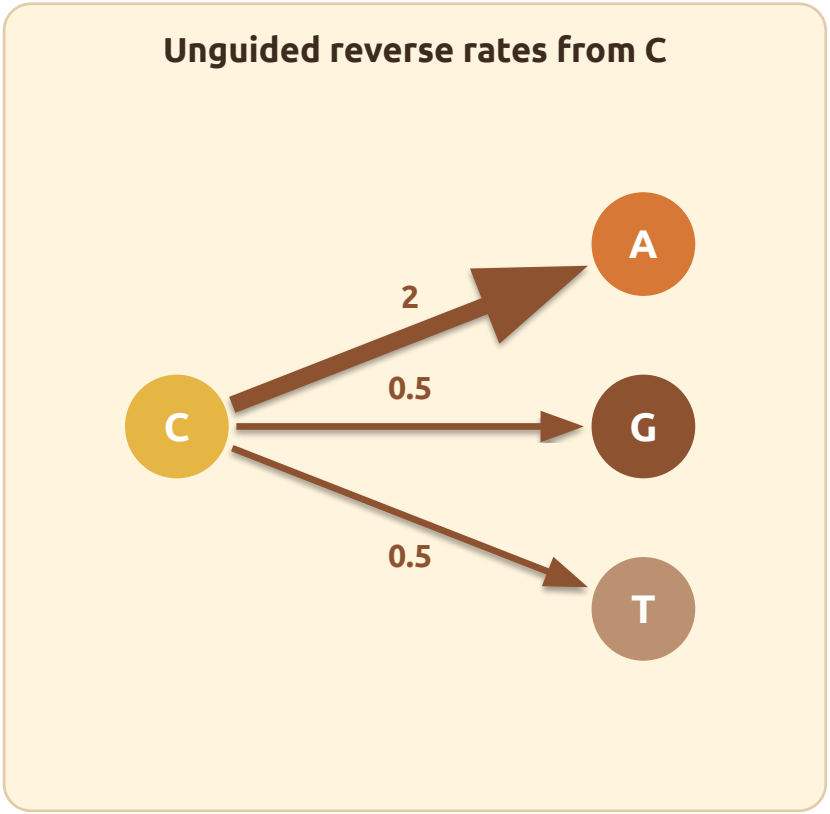


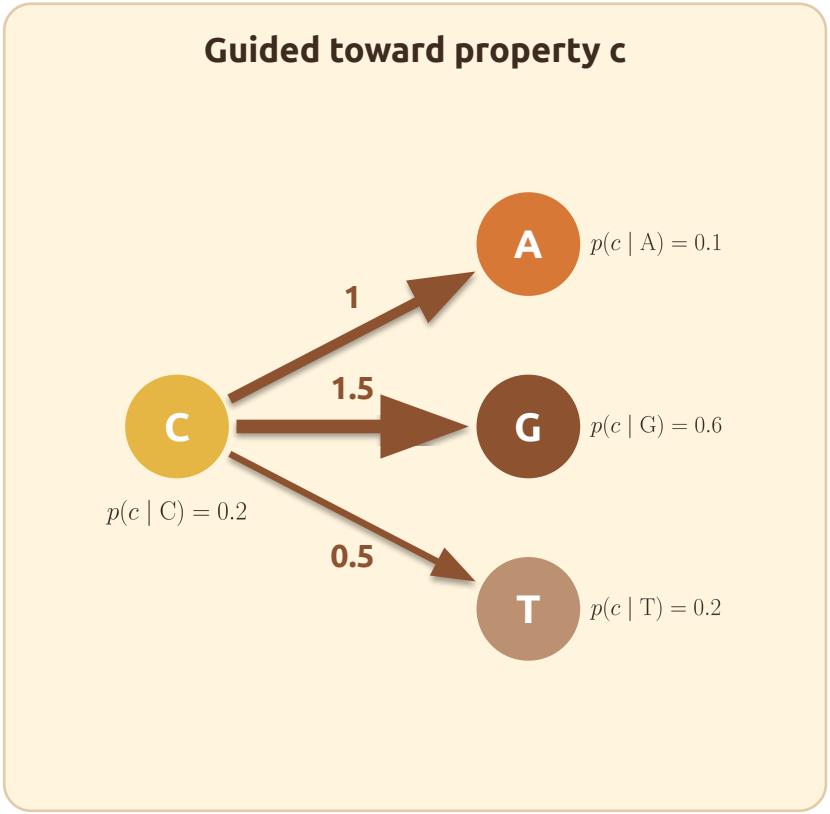
The rate toward G triples, because p(c) rises from 0.2 to 0.6





With guidance, we reweight each jump by the change in h





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$$\mathrm{KL} = \sum_{y \neq x} hA(y) \log \frac{A(y)}{B(y)} \; + \; (1 - hA_{\mathrm{out}}) \log \frac{1 - hA_{\mathrm{out}}}{1 - hB_{\mathrm{out}}} + o(h)$$

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The KL divergence over one short step equals the sum over destinations y other than x of the jump probability hA(y) times the log of A(y) divided by B(y), plus the staying probability $1 - hA_{out}$ times the log of $1 - hA_{out}$ divided by $1 - hB_{out}$, up to o(h).

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For small h, log(1 – u) ≈ –u, so we can expand the staying term:

$$(1 - hA_{\mathrm{out}}) \log \frac{1 - hA_{\mathrm{out}}}{1 - hB_{\mathrm{out}}} = h\,(B_{\mathrm{out}} - A_{\mathrm{out}}) + o(h)$$

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For small h, the staying term equals h times B_{out} minus A_{out}, up to o(h).

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A_{out} and B_{out} are the total target and model rates of leaving x.

Next, we go from one short step to a whole path

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Divide by h, add up along the path, and integrate over time:

$$\mathrm{KL}(\text{paths}) = \mathbb{E} \int_0^1 \sum_{y \neq X_t} \left[A \log \frac{A}{B} - A + B \right] dt$$

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The KL divergence between the target and model path distributions equals the expected integral over time t from 0 to 1 of the sum over destinations y other than X_t of A times the log of A divided by B, minus A, plus B, where A is the target rate and B is the model rate of the jump to y.

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Without the staying term, extra jumps would cost nothing, and with the term B – A, we penalize every extra jump.

Look at one term of this sum

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The rate error ℓ(A, B) between a target rate A and a model rate B equals A times the log of A divided by B, minus A, plus B.

Look at one term of this sum

For one destination, define $\ell(A, B) = A \log \frac{A}{B} - A + B$

Use log u ≤ u – 1 with u = B/A, so ℓ(A, B) ≥ 0, with equality exactly when B = A:

$$A \log \frac{A}{B} = -A \log \frac{B}{A} \geq -A\left(\frac{B}{A} - 1\right) = A - B$$

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A times the log of A over B equals negative A times the log of B over A, which is at least negative A times B over A minus one, and that equals A minus B.

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Now let the target A be uncertain and minimize the expected error over B:

$$\boxed{\frac{\partial}{\partial B}\,\mathbb{E}\big[\ell(A, B)\big] = 1 - \frac{\mathbb{E}[A]}{B} = 0 \quad \Longrightarrow \quad B^* = \mathbb{E}[A]}$$

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The derivative of the expected rate error E[ℓ(A, B)] with respect to the model rate B equals one minus the expected target rate E[A] divided by B, so the optimal rate B^* equals E[A].

Look at one term of this sum

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Use log u ≤ u − 1 with u = B/A, so ℓ(A, B) ≥ 0, with equality exactly when B = A:

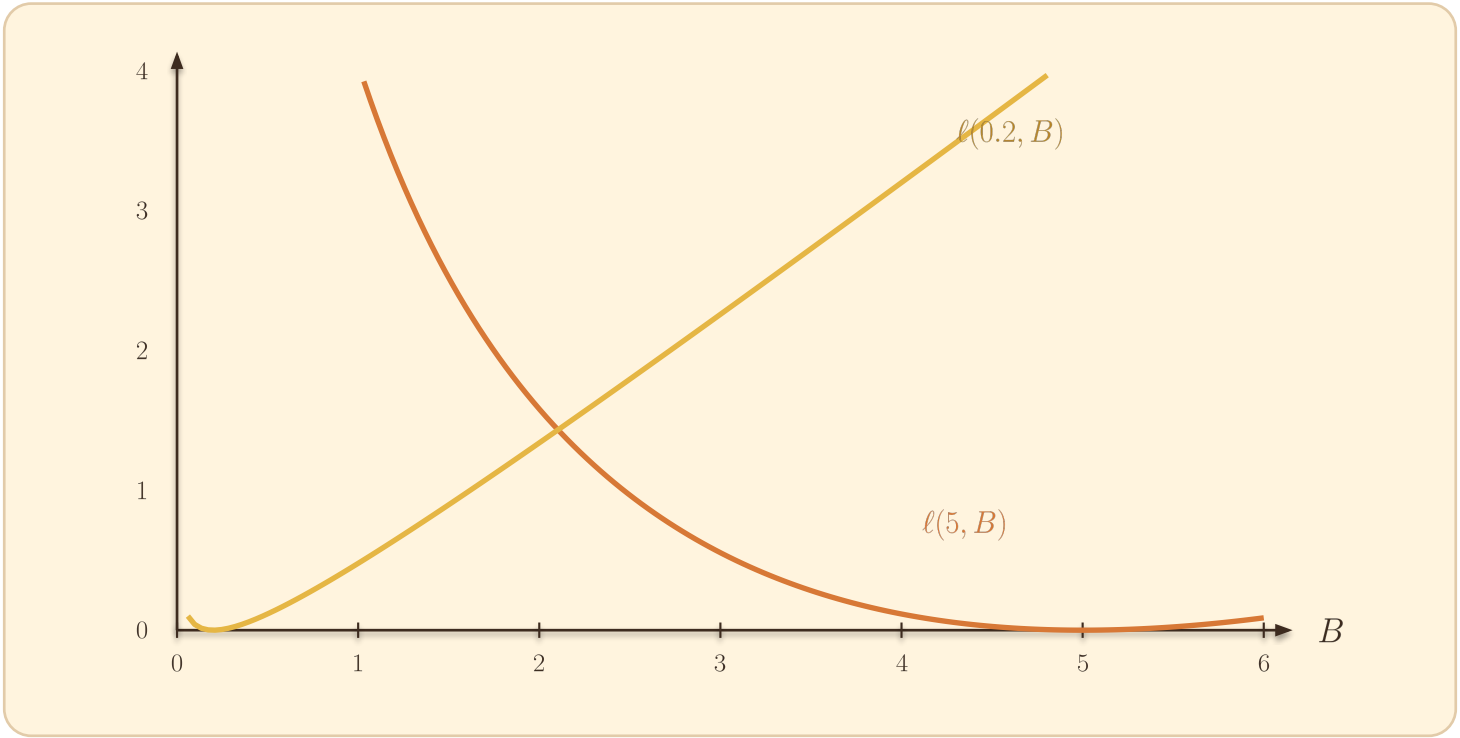
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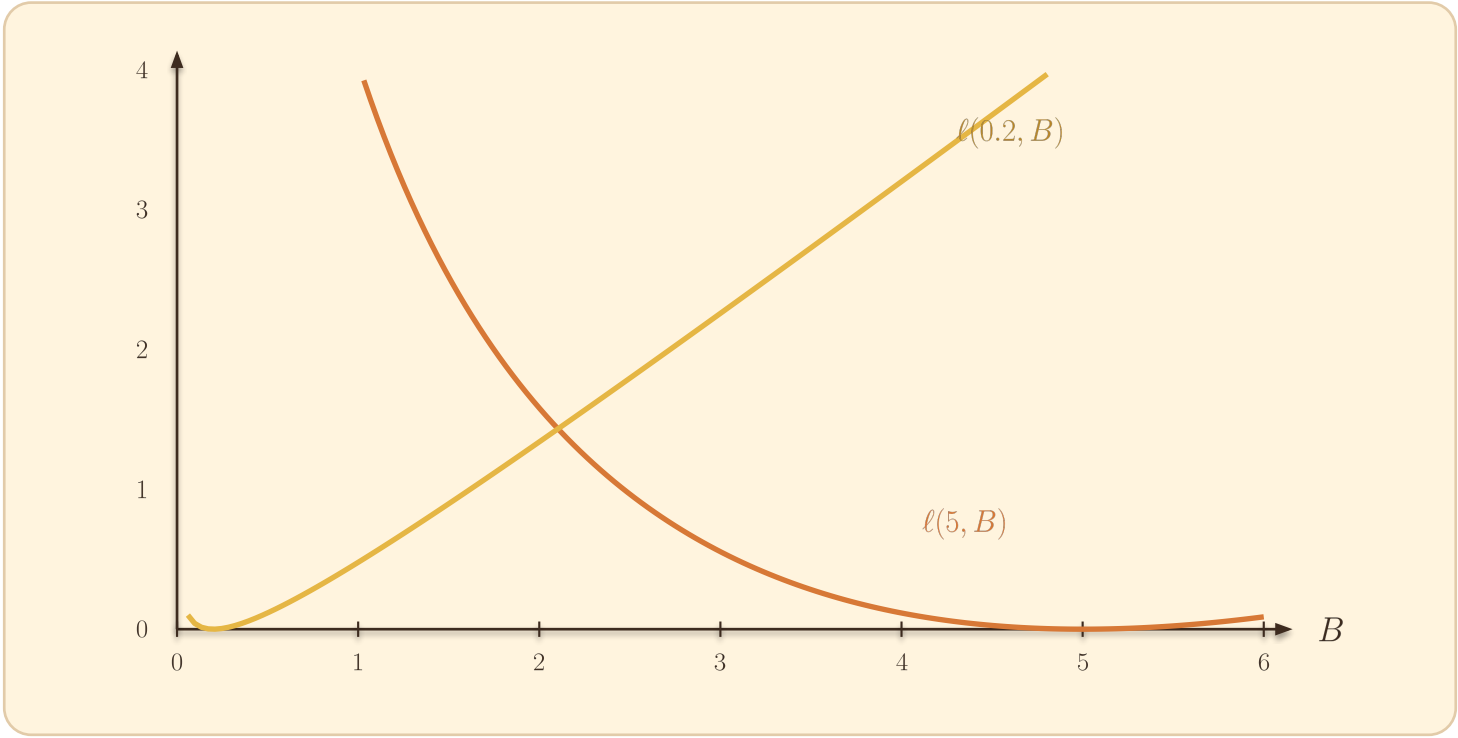
$$\boxed{\frac{\partial}{\partial B}\,\mathbb{E}\big[\ell(A, B)\big] = 1 - \frac{\mathbb{E}[A]}{B} = 0 \quad \Longrightarrow \quad B^* = \mathbb{E}[A]}$$

In Lecture 2, with squared error we recovered the conditional mean of the velocity. Here, with the rate error we recover the conditional mean of the reverse rate.

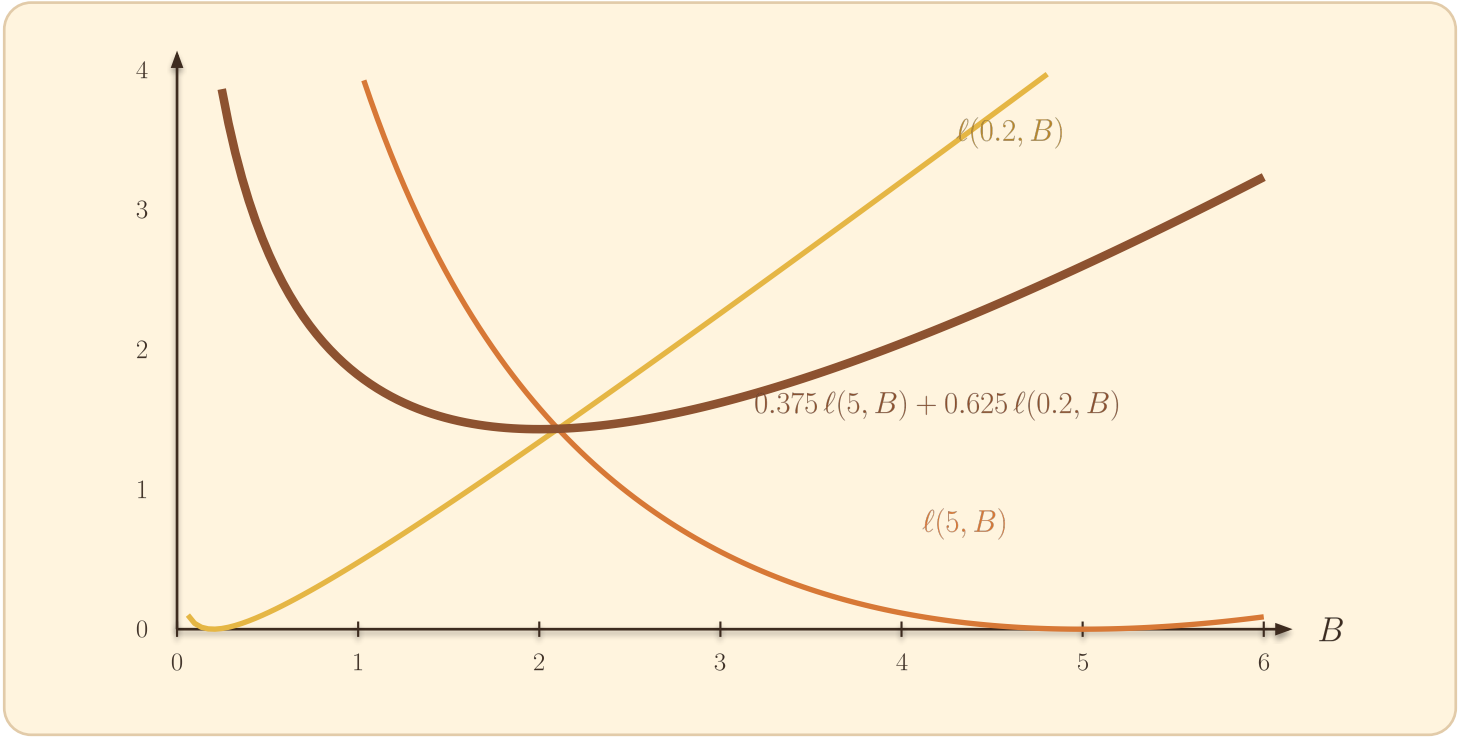
Let’s plot that error for our two conditional targets



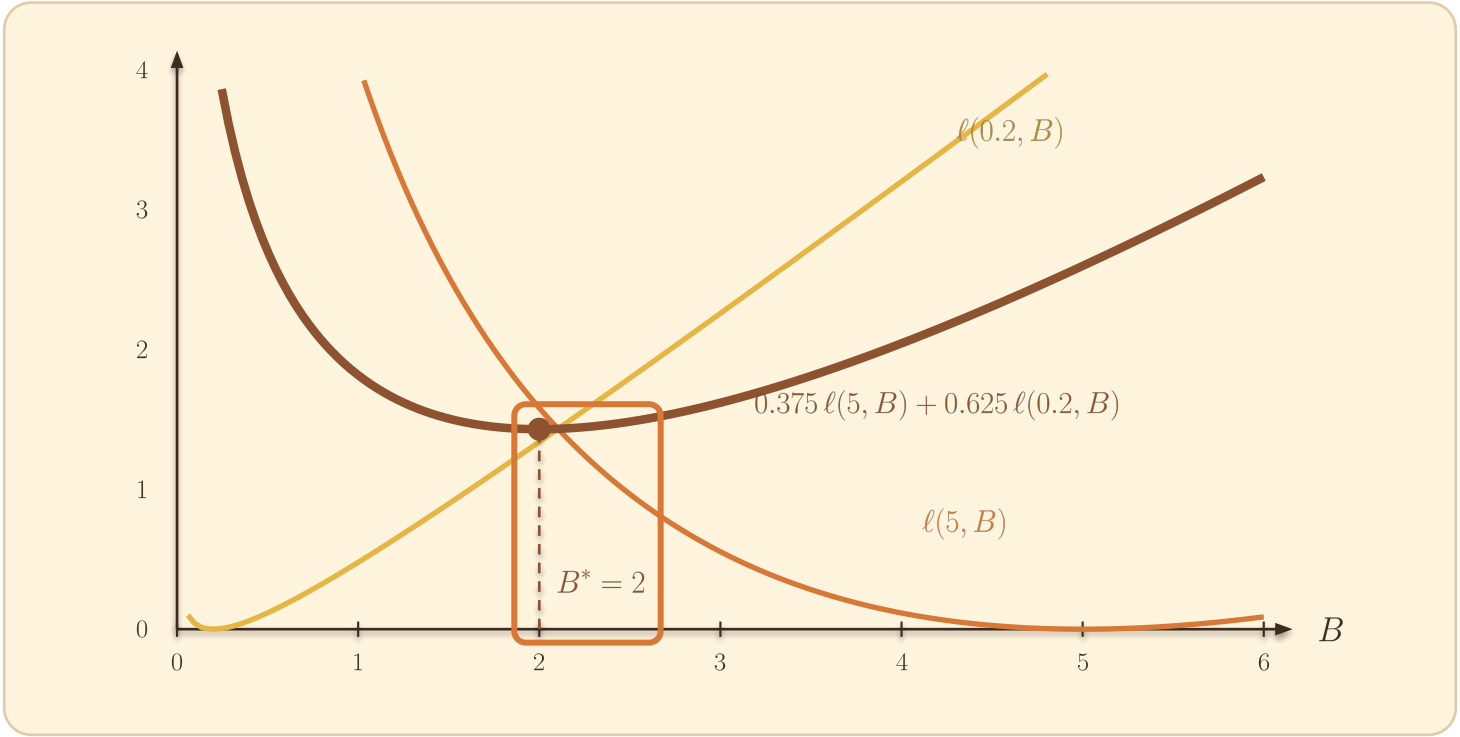
Each curve is smallest at its own target rate, 5 and 0.2



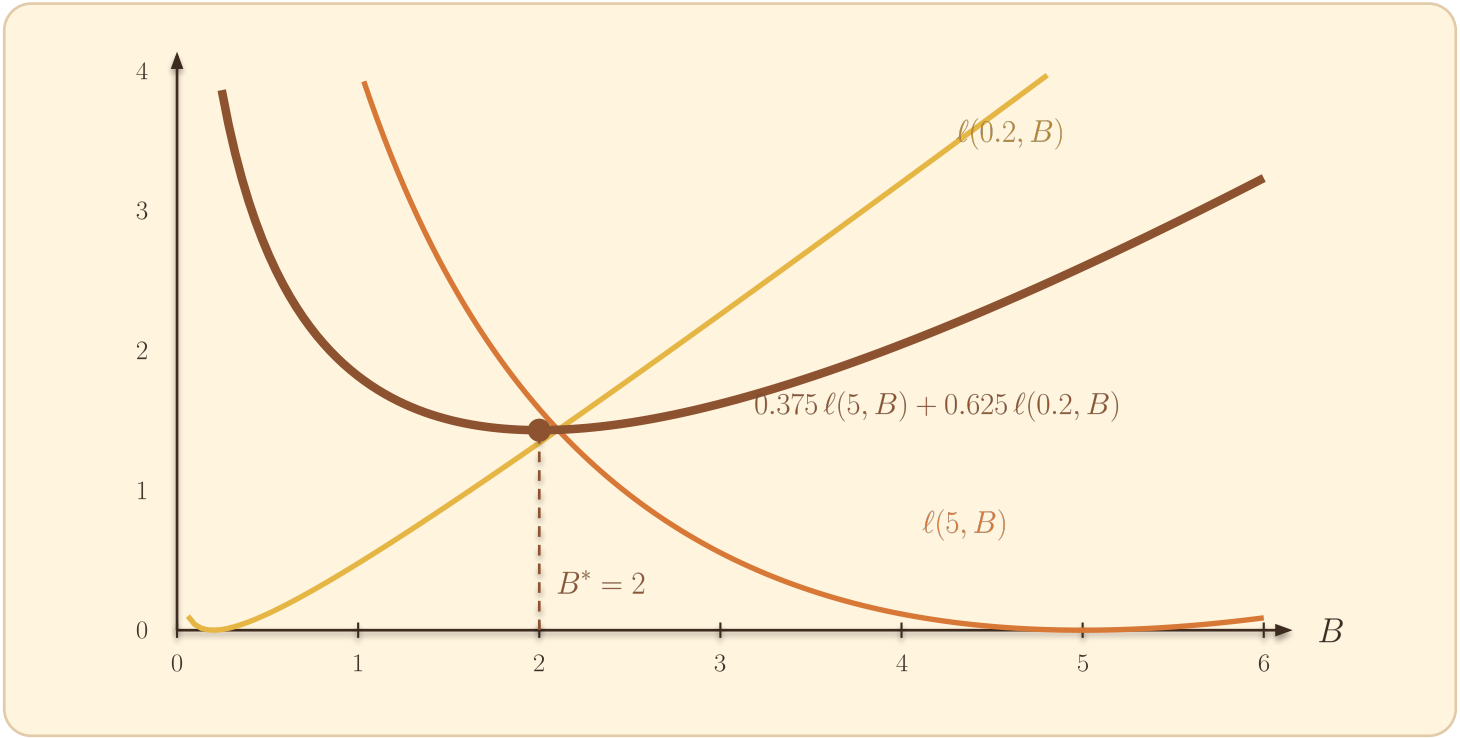
Now weight them by the posterior to get the error we actually minimize



Its minimum sits at the posterior-average rate, 2



So when we train on conditional targets, we get the marginal reverse rate



Suppose we could evaluate the marginal reverse rates

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In the ideal loss, we compare the model rates with the marginal reverse rates:

$$\mathcal{L}_{\text{marg}}(\theta) = \int_0^1 \mathbb{E}_{Z_t} \sum_y \ell\big(\hat{R}_t(Z_t, y),\ \hat{R}_t^\theta(Z_t, y)\big)\, dt$$

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The marginal loss $L_{marg}(\theta)$ equals the integral over time t from 0 to 1 of the expectation over noisy states Z_t of the sum over destinations y of the rate error ℓ between the marginal reverse rate $\hat{R}_t(Z_t, y)$ and the model rate $\hat{R}^\theta_t(Z_t, y)$.

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$$\mathcal{L}_{\text{cond}}(\theta) = \int_0^1 \mathbb{E}_{x_0,\, Z_t} \sum_y \ell\big(\hat{R}_t(Z_t, y \mid x_0),\ \hat{R}_t^\theta(Z_t, y)\big)\, dt$$

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The conditional loss $L_{cond}(\theta)$ is the same integral with an expectation over clean sequences x_0 and noisy states Z_t, and with the conditional reverse rate $\hat{R}_t(Z_t, y \mid x_0)$ as the target for the model rate $\hat{R}^\theta_t(Z_t, y)$.

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For each Z_t, the posterior average of the conditional targets is the marginal rate, so both losses have the same minimizer and the same gradient.

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Each clean training sequence supplies a tractable target; averaging across examples teaches the rate needed at the noisy state.

Conditional Rate-Matching Theorem

In the ideal loss, we compare the model rates with the marginal reverse rates:

$$\mathcal{L}_{\text{marg}}(\theta) = \int_0^1 \mathbb{E}_{Z_t} \sum_y \ell\big(\hat{R}_t(Z_t, y),\ \hat{R}_t^\theta(Z_t, y)\big)\, dt$$

We cannot evaluate $\hat{R}_t$, because it averages over every clean sequence that could have produced Z_t. We can evaluate the conditional rates for each training sequence:

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For each Z_t, the posterior average of the conditional targets is the marginal rate, so both losses have the same minimizer and the same gradient.

Next, we relate this loss to the likelihood

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Add the prior term at t = 1:

$$\boxed{-\log p_\theta(x_0) \le \mathcal{L}_{\text{cond}}(x_0;\theta) + D_{\text{KL}}\big(q_1(\cdot \mid x_0) \,\|\, p_1\big)}$$

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The negative log-likelihood of a clean sequence x_0, negative log $p_\theta(x_0)$, is at most its conditional rate loss $L_{cond}(x_0; \theta)$ plus the KL divergence between $q_1(\cdot \mid x_0)$, the fully noised distribution of x_0, and the prior p_1.

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This continuous-time bound was derived in Campbell et al. (NeurIPS 2022), and in SEDD (ICML 2024) it is written in terms of ratios.

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A low training bound forces a low negative log-likelihood; a high bound may simply be loose.

Now build the model rates from the denoiser

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Use the denoiser in place of the posterior in the reverse-rate formula:

$$\hat{R}_t^\theta\big(z, z^{i\leftarrow a}\big) = R_t\big(z^{i\leftarrow a}, z\big) \sum_b d_\theta^i(b \mid z, t)\, \frac{q_t(a \mid b)}{q_t(z^i \mid b)}$$

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The model rate $\hat{R}^{\theta}_t(z, z^{i\leftarrow a})$ for changing position i of z to token a equals the forward rate $R_t(z^{i\leftarrow a}, z)$ back from $z^{i\leftarrow a}$ to z, times the sum over clean tokens b of the denoiser probability $d^i_\theta(b \mid z, t)$ times $q_t(a \mid b)$ divided by $q_t(z^i \mid b)$.

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Every model rate is nonnegative, and when the denoiser matches the posterior, every model rate equals the marginal reverse rate.

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Every model rate is nonnegative, and when the denoiser matches the posterior, every model rate equals the marginal reverse rate.

The network outputs one categorical distribution per position, which is the denoiser we already trained for MDLM.

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The network predicts plausible clean tokens, and the known corruption process turns those probabilities into jump rates.

With rates in hand, we generate with short reverse steps

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Over a short step h, each position jumps independently with probability proportional to its reverse rate:

$$\Pr\left(Z^i_{t-h} = a \mid Z_t = z\right) \approx h\, \hat{R}^\theta_t\left(z, z^{i \leftarrow a}\right), \qquad a \neq z^i$$

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The probability that position i holds token a at the earlier time t – h, given the sequence z at time t, is approximately h times the model rate $\hat{R}^\theta_t(z, z^{i \leftarrow a})$, for every token a other than z^i.

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Each step is valid when h times every position's total reverse rate is at most one, the same condition we found for the forward step.

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Several positions can change within one step. This approximation is called τ-leaping, and it becomes exact as h shrinks.

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Several positions can change within one step. This approximation is called τ-leaping, and it becomes exact as h shrinks.

For the π family we can also take an exact posterior step over any interval, as we did for MDLM.

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Several positions can change within one step. This approximation is called τ-leaping, and it becomes exact as h shrinks.

Larger steps save model calls, but the rates can become outdated after another position changes.

Now add a corrector step at a fixed time

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The corrector rate matrix C_t equals the forward rate matrix R_t plus the reverse rate matrix $\hat{R}_t$, both taken at the same time t.

Now add a corrector step at a fixed time

We can reduce accumulated sampling errors by mixing at a fixed time without disturbing p_t.

Add the forward and reverse rates at a fixed time t: $C_t = R_t + \hat{R}_t$

By the reversal identity, the two terms exchange places, so this process is in detailed balance with p_t:

$$p_t(x)\, C_t(x, y) = p_t(y)\, C_t(y, x)$$

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The probability $p_t(x)$ times the corrector rate $C_t(x, y)$ from x to y equals the probability $p_t(y)$ times the corrector rate $C_t(y, x)$ from y back to x.

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So p_t is stationary for C_t: when we run C_t at a fixed time, the correct marginal stays in place and earlier errors mix toward it.

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Predictor steps move through time, and corrector steps stay at one time (Campbell et al., NeurIPS 2022). We will use this idea in P2.

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A corrector revisits token choices while keeping the target noise level fixed.